The Rubik's Cube Group

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1 Opening

The first thing people think about Rubik's cube is that they're all about maths and calculations. As anyone that can solve one will say, this is far from the truth. People have been solving the Rubik's cube and other twisty style puzzles for many, many years, with the world record being just over 3 seconds.

But why has the cube become so associated with maths? Is the maths behind it even useful? To answer this, let's rewind back to the 1970s...

1.1 Conception

It's 1974: ABBA's 'Waterloo' just won the Eurovision song contest, and Dungeons & Dragons was first released. A Hungarian architect professor Ernő Rubik is exploring how to construct a building that has moving rooms within 3D space without falling apart. He came up with a design, and after colouring the sides of the pieces, realised he'd created an interested puzzle.

He initially patented his 'Magic Cube' in Hungary 1975, then later renamed it to the 'Rubik's Cube' for trademark protection before it was distributed outside of Hungary.

Up until today, over 500 million cubes have been sold around the world, that's not including all of the other twisty style puzzles produced as well!

1.2 Singmaster

One purchaser of the puzzle was American-British mathematician David Singmaster.

Singmaster was attending the International Congress of Mathematicians held in August 1978. There, notable mathematicians John Conway and Roger Penrose were playing with the new 'Hungarian Magic Cube', and discussing their solutions.

> "Penrose did not believe anyone could do it without consulting a pocket full of notes, and his coat pocket was full of envelopes with move diagrams written on them." - Singmaster, 1978

Singmaster was able to obtain a cube at the event - in exchange for an M. C. Escher book - and began creating his own solution from overhearing the conversation between Conway and Penrose.

Singmaster later wrote one of the first articles about the cube in 1979, and later in the same year, his 'Notes on the "Magic Cube"' which contained all of his mathematical discoveries of the cube. He invented his own 'Singmaster' notation for puzzle, which is still used today, as well as the 'Layer by Layer' method, which is commonly taught as a beginners method for the cube.

This video we'll dive into Singmaster's analysis of the puzzle, and how he managed to use ideas of group theory the pioneer the puzzle.

2 Observations

The first thing any mathematician should do before approaching a difficult puzzle is to closely analyse the properties and constraints set by the puzzle, rather than blindly turning the cube and hoping it solves.

Disassembling the puzzle, we see we have 4 different types of pieces: centre pieces, edge pieces, corner pieces, and the core.

- The core simply attaches to each of the centre pieces, such that all of the centres spin in place. This means the centre pieces do not move relative to each other (since they're physically connected to each other), so the centre pieces actually tell us what colour that side should be, and they never mix up.
- The edge pieces have two coloured faces, since they lie between two centre pieces. Because turns of puzzle can only be 90°, this means edges always lie between two centre pieces, meaning edges always act as edges.
- Similarly, the corners lie between 3 centres and hence have 3 colours. Like the edges, corners will also always act as corners, so corners and edges (and centres) will never swap with each other.

To help us describe and write down movements we perform on the puzzle, Singmaster created a useful notation for the Rubik's Cube. He gave each face of the cube a letter, depending on where the user is facing the puzzle: U, D, L, R, F, B, for the up, down, left, right, front and back faces. A letter on its own means to turn that face 90°clockwise. An apostrophe (or prime) after the letter, like R', means to rotate that face anti-clockwise/counterclockwise. And a 2 after the letter, like U2, means to rotate that face 180°(notice that a 180°rotation is the same applied clockwise or anti-clockwise). Strings of these moves together form what we call an algorithm.

If we turn the puzzle, such as a U turn, and watch the edge pieces really closely, we see they actually cycle about in a circular fashion: either clockwise or anticlockwise. This is equivalent to laying the edges out on the table, and cycling them as such, which simply rearranges the order of the edges.

This looks remarkably like something called a permutation in mathematics. Let's look further into them and see what we can find out.

3 Permutations

The mathematical definition of a permutation is 'a bijection of an ordered set onto itself'. We'll see where this definition comes from by looking at a simple example of boxes with balls inside.

3.1 Balls and Boxes

Let us have 4 boxes, labelled A, B, C and D, each with a correspondingly named ball inside. We want to look at the different ways we can place each ball into each box, assuming a box must contain one and only one ball.

We can represent this with a function $\mathcal{P}$. We can let the input of the function be the box we want to look at, and it'll output which ball is inside of it. For instance,

$$\mathcal{P}(A) = C, \mathcal{P}(B) = B, \mathcal{P}(C) = D, \mathcal{P}(D) = A$$

This is obviously quite abstract, so a better way to represent this is to assume each ball starts in its corresponding box, and track each ball's travel to their new box. To make it even more clear, we can have two sets of boxes, and draw arrows from the first set of boxes to the box in the other set the ball travels to, for each ball of course.

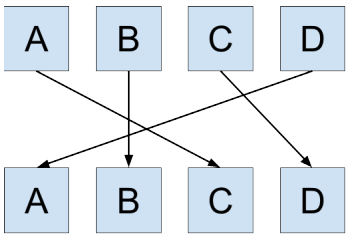


Figure 1: Simple 4 item permutation, denoted $\mathcal{P}$.

Figure 1 is a single permutation of our balls. There are obviously many different permutations these balls can be in, 24 to be exact. We have a special name for one of these: the identity permutation, which we often denote as $\mathcal{I}$. This is the permutation equivalent to moving no balls (aka each ball maps to its own box).

With this notation, we can start to string together permutations, or 'composing' permutations. We can simplify this composition by rewriting it out as a single composition, as in Figure 2. If we denote the first permutation $\mathcal{A}$ and the second $\mathcal{B}$, then we write the composition of them as if we're multiplying them, so the composition is simply $\mathcal{AB}$[1].

3.2 Number of Permutations

One useful fact to understand is how to determine how many permutations there are for a finite number of objects. Let's look at 4 objects again and then extrapolate our result.

Begin with no balls in any boxes. We'll place one ball at a time in a box, and count how many choices we can make picking which boxes to put the balls in.

[1]In mathematics, composition is normally written right-to-left due to the convention of how we write functions. Despite this, all further compositions will be written left-to-right to avoid confusion; all of the maths works the exact same.

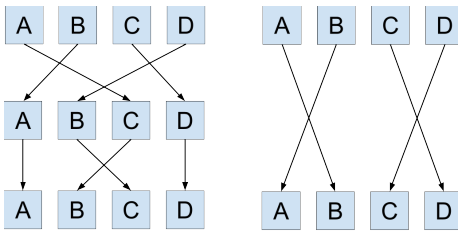


Figure 2: Composition of 2 permutations, and its simplified form.

- Initially all of the boxes are empty, and so the first ball can go in any one of the four boxes.
- For the second ball we have a choice of 3 boxes, since one of them now has the first ball inside. So for 2 balls we have $4 \times 3 = 12$ possible arrangements they could be in.
- The third ball now only has two boxes left to go inside. So our number of permutations doubles because we have 2 choices, and hence we get $4 \times 3 \times 2 = 24$ choices for the 3 balls.
- The last ball has only one remaining box it can go in this is the same as multiplying our number by 1, which does nothing to our answer.

Hence the number of permutations of 4 balls is $4 \times 3 \times 2 \times 1 = 24$. In general we can see that the number of permutations of n balls is just

$$n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1 = n!$$

We often denote this n!, where ! is the **factorial** function. It just multiplies n by every positive number smaller or equal to it. This will be useful to understand later.

3.3 Parity

Something important that we should keep in mind is the '**crossing number**' of a permutation, which is simply the number of times two arrow lines cross. In Figure 2 the first permutation's crossing number is 3, the second is 1, so it might seem logical that the crossing number of the permutation as a whole is 4. However, 'untangling' or simplifying the permutation, we see that it's actually 2. How come?

Observe that to 'untangle' the arrows/strings, we find any two lines that have a double crossing and swap them over. This is because if a line double crosses another, then the lines didn't need to cross in the first place. Hence untangling means we remove these double crossings. When removing a double crossing, the total crossing number decreases by 2. So untangling the lines will always decrease the crossing number by a multiple of 2 (i.e. an even number).

This is useful information since we know

- Odd - **Even** = Odd
- Even - **Even** = Even

And so the even/odd-ness of the crossing number remains the same after untangling. Mathematically we call this even/odd-ness characteristic the **parity** of the permutation. We'll call a permutation 'even' if it has even parity (an even crossing number), and 'odd' if it has odd parity (an odd crossing number). We'll come back to parity later.

3.4 Order

We notice something interesting when we compose the same permutation with itself multiple times. If we compose Figure 1 with itself 3 times, written as $\mathcal{P}^3$, we notice we get the identity permutation (the permutation that moves nothing). In fact, any (finite) permutation has this property.

We can see this since for any individual mapping in a permutation, either the ball maps back to its own box (in which we're done), or it maps to a different box. Since there are finitely many boxes eventually the ball must get back to where it started, since eventually we'll run out of boxes. This is true for each individual ball.

We'll call the smallest number of times we must compose a permutation with itself to get to the identity permutation the 'order' of the permutation. So Figure 1 has order 3. Repeating this 6, 9, 12, ... many times also results in the identity, as these are all multiples of 3.

3.5 Cycle Notation

Our current method of showing permutations is nice and intuitive, but as we increase the number of balls and boxes, clearly our diagrams start to become very complicated and lengthy to draw. Instead we resort to new forms of notation.

3.5.1 Two-Line Notation

Instead of keeping the same boxes in front of each other, let's shuffle the boxes such that the arrows to each box all become parallel to each other. Since now we know all of the arrows are parallel, we can scrap them completely, and simply write Figure 1 as,

$$\begin{pmatrix}1234\\3241\end{pmatrix}$$

This is **two-line notation**. This notation is useful, as there may be no natural ordering of our boxes in different contexts. There's a natural ordering to our balls and boxes since we labeled them in a natural order. However, pieces on a Rubik's Cube don't have a natural order (what's to say the white-blue edge comes 'before' the orange-green edge?).

3.5.2 Cycle Notation

Another notation which also avoids this problem, which we'll use a lot, is **cyclic notation**. Like we described in the 3.3 Order section, a single ball can be tracked , and will travel to anywhere from 1 to N boxes, where N is the order of the permutation. We can do this such that we have a ball mapping to every single box in our permutation.

Start with ball 1 in Figure 1. We see ball $1 \mapsto 3 \mapsto 4 \mapsto 1$. The only box ball 1 doesn't visit is box 2, which we can clearly see maps to itself. In cyclic notation we express this as (134)(2). Here we see our Figure 1 permutation is made of two **disjoint** cycles. Disjoint means that the boxes referenced in each cycle don't contain the same number(s), and hence act as two separate cycles, in our case the mapping of box 2 to itself, (2), doesn't interact with the cycle (134), hence they're disjoint.

We can write out more complicated cycles, like (2 4 5 3 1)(6 7 9 8). Notice cycle notation is not unique: the order disjoint cycles are written does not matter, so (6 7 9 8)(2 4 5 3 1) represents exactly the same cycle as the one above. Also, since cycles are cyclic, each individual cycle can start at any point in the cycle, as the structure is preserved, i.e. (6 7 9 8) $\iff$ (7 9 8 6) $\iff$ (9867) $\iff$ (8679) are all identical cycles.

We call the given cycle (6798) a 4-cycle, because clearly if we repeat this cycle 4 times, each ball will end up where they started. Likewise a 3-cycle like (257) has order 3 as it cycles the balls 2, 5 and 7. A 2-cycle (24) is equivalent to just swapping two balls, hence they must have order 2. And single cycles/1-cycles like (2) have order 1, meaning they do nothing.

It's possible to combine to cycles that are not disjoint, for instance (123)(12). To simplify this into disjoint cycles, for each number that appears, we track the movement each takes from the left-most cycle to the right-most cycle. Starting with 1: 1 goes to 2 in the first cycle, then 2 goes back to 1 in the second cycle. Next: 2 goes to 3 in the first, and that's it since 3 doesn't appear anywhere else. Lastly: 3 goes to 1, and 1 then goes to 2. Hence our simplified cycle becomes (1)(23).

Order gets slightly more complicated with disjoint cycles, like (123)(4567). Clearly this permutation consists of a 3-cycle and a 4-cycle which don't affect each other. So for any 3 times we repeat this permutation, the 3-cycle becomes the identity permutation and we're left with the 4-cycle. Likewise, repeating the permutation 4 times results in the 4-cycle becoming the identity, and we're left with the 3-cycle. So to get the entire permutation to become the identity, we must apply it to itself by a number that's both a multiple of 3 and 4. The lowest number satisfying this will be the order of the permutation, which is simply the lowest common multiple of 3 and 4. In this case $lcm(3, 4) = 12$, hence the order of (123)(4567) is 12.

So a permutation made up of different cycle lengths, then the order of that permutation will be the lowest common multiple of all of those cycle lengths.

The parity of a cycle can be easily determined, since any cycle can be broken down into (not necessarily disjoint) 2-cycles[2]. For example,

[2]It's not necessarily true that, for example, a 3-cycle can **always** be expressed as an even number of swaps, but we can show with more rigour, that this will always be the case for all cycles.

$$(123) = (12)(13)$$

$$(1234) = (12)(13)(14)$$

And so on.

A 2-cycle always has odd parity (think back to crossing-numbers), so the parity of any cycle is only dependent on the size of the cycle. If the cycle is amongst an even number of elements, the parity is odd, and vice versa. Since permutations can be expressed as disjoint cycles, then we can easily find the parity of these as well by finding out how many 2-cycles they're comprised of.

3.6 Application to the Rubik's Cube

Now using our built up knowledge of permutations, let's apply these to the Rubik's Cube.

First we need a way to identify every piece on the cube. We could use the colours of each of the pieces, like blue-red, green-white-orange, but if we're given a mixed up cube, these pieces could be anywhere on the puzzle, which is not helpful if we want to create a general solution to the puzzle.

Instead, let's identify each pieces by it's position on the puzzle. We associate each face with a letter: R, L, U, D, F, B. We can describe an edge piece by the two faces it lies between, for example the edge directly in front of the user is the UF edge (or equivalently the FU edge), likewise corner pieces are described by the 3 faces they lie between, like the UFR corner.

Now if we observe the turning of a single face, like applying a U turn, we can see the edges and corners perform this permutation

$$(UF, UL, UB, UR)(UFR, UFL, UBL, UBR)$$

In fact, we can write any set of moves we apply to the cube using cycle notation. For instance the moves R2 U2 is equivalent to the permutation

$$(FR, BR)(UF, UB)(UR, UL, DR)(UFR, UBL, DBR)(UFL, DFR, UBR)$$

We can use what we learnt about the order of permutations and can see that if we apply the algorithm R2 U2 three times, then the 3-cycles will become the identity, and we'll be left with just the two sets of edge swaps. This is exactly what we see when we apply this on the cube.

3.6.1 Notation Limitations

We give another example of the algorithm R U on the puzzle. The algorithm can be expressed as the permutation:

$$(UF, UL, UB, UR, BR, DR, FR)(UFL, ULB, URB, RDB, FDR)$$

Which is the product of a disjoint 5-cycle and 7-cycle. Hence from what we learnt before, the permutation should have order $lcm(5, 7) = 35$. However, performing this algorithm 35 times doesn't get us back to the solved state. Why?

We notice that this state after 35 iterations actually leaves all of the pieces in their correct positions, but the corners are twisted in place. This is our limitation: our notation doesn't account for the rotation/orientation of the pieces.

The best way to mitigate this is to assign a number to each of the 48 different stickers to the puzzle, and rewrite our permutations using these numbers.

So according to Figure 3, a U turn is now written:

$$(17\,19\,24\,22)(6\,25\,43\,16)(8\,30\,41\,11)(7\,28\,42\,13)(18\,28\,42\,13)$$

The rest of this video can be thought about as this labelling of every single sticker on the cube. However, it becomes difficult to tell, using this notation, which pieces are moving where. So while we could continue using this new notation, we'll actually switch back to our old one for clarity sake, with a few changes.

Mainly, if an edge, say the UF edge, remains in place but flipped, we denote that with (UF) alone. If a corner, say the UFR corner, is twisted in place, we denote that with (UFR), where the sticker that was on the U face goes to the F face, the one on the F face goes to the R face and so on. So $(RU)^{35}$ becomes (UFR)(ULF)(UBL)(URB)(RDF)(RDB).

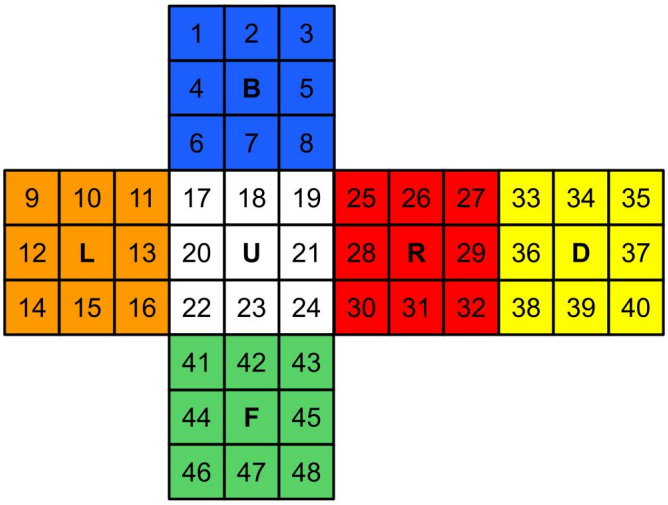


Figure 3: A possible labelling of the stickers of a cube.

4 Groups

We have learnt everything we need to know about permutations and how they relate to the Rubik's Cube. Permutations have a very nice relation to an area of maths called **Group Theory**. We're going to explore what a group actually is, what permutation groups are, and why it even matters to know the Rubik's cube is in fact a group.

4.1 What is a Group?

In simple terms, a group is just a set of things with certain properties.

A set can be any list of things, like: the whole numbers, the spoons in your house, all of the ginger cats living in the UK, every single atom in my body... We call each individual thing an element of that set.

We'll denote some arbitrary set of objects as $\mathcal{G}$. To define a group, we need to define a binary operation between any two elements in our set. 'Binary' just means an operation that takes two inputs and spits out a single output: addition is a simple binary operation. I'll denote an arbitrary operation with $*$.

Now we enforce 4 rules our set and operation $(\mathcal{G}, *)$ must follow, called the group axioms.

1. Our operation * must be **closed** in $\mathcal{G}$. This means for any two elements a and b in $\mathcal{G}$, $a * b$ must also be an element of $\mathcal{G}$.

2. Our operation must be **associative** in $\mathcal{G}$. This means rearranging the brackets/parentheses in an expression will not change the result, meaning for a, b, c in $\mathcal{G}$, $a * (b * c) = (a * b) * c$.

3. Our set $\mathcal{G}$ must contain a special element, e, called the **identity** element. This element has the property that it leaves any element the same when operated on, so for any a in $\mathcal{G}$, $a * e = e * a = a$.

4. Every element in $\mathcal{G}$ has an **inverse** element, such that an element operated with its inverse gives us the identity element. In maths terms, for an element a in $\mathcal{G}$, there exists some element a^{-1} in $\mathcal{G}$ such that $a * a^{-1} = a^{-1} * a = e$.

4.2 Examples of Groups

We're going to look at a two simple examples of groups.

4.2.1 The Group of Integers with Addition

Let's take the set of whole numbers, or integers, that is including zero and all of the negative integers. This set is often denoted as $\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$. We'll define our binary operator as just addition, $+$.

How do we know this forms a group? Let's check the axioms.

1. Clearly if we add two integers together, we get another integer. Hence our set is closed under addition.
2. Addition is clearly associative. Try out the definition for any three integers. (We don't give a full proof for this as it's a bit long and not necessary).
3. You can probably guess what the additive identity is (we want an integer such that for any other integer a that $a + e = e + a = e$), it's just zero.
4. $\mathbb{Z}$ contains the negative of any integer we choose from it (if it's negative, then the negative of a negative is positive, and obviously $0 = -0$), hence every element in $\mathbb{Z}$ has an inverse.

So we know that $(\mathbb{Z}, +)$ forms a group!

4.2.2 The Group of Triangle Transformations

Another group is all of the ways we can transform a triangle (or in fact any regular polygon). When I mean transform, I mean transform the triangle such that it still remains as the same triangle as we started with. This means we're limited to just rotations and reflections of the triangle.

We can rotate the triangle in 60°increments to maintain the triangle's initial shape. Let's denote a single 60°clockwise rotation of the triangle with r. For a 120°rotation we'll denote with r^2, as it's like applying r twice. Once more rotation gets us back to where we started, the identity transformation, that we'll denote with e.

There are 3 mirror lines the triangle can be reflected across. Let's denote each mirror line as m_1, m_2, m_3. Then each associated reflection will just be represented by its mirror line.

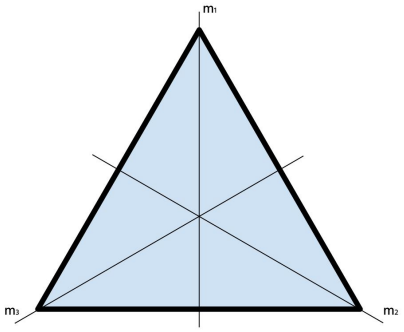


Figure 4: The mirror lines of a triangle.

I propose that composing these transformation operations together forms a group. A way to show this is to write every combination of transformations in a table, often called a Cayley table. Unlike the group of integers with addition, we write composition of elements as if we're multiplying numbers in algebra, omitting the $*$ symbol. Since in future we'll be working with permutation groups under composition, from now on I'll use multiplication notation for group operations, but everything can equally be done using $*$ notation.

$*$	e	r	r^2	m_1	m_2	m_3
e	e	r	r^2	m_1	m_2	m_3
r	r	r^2	e	m_2	m_3	m_1
r^2	r^2	e	r	m_3	m_1	m_2
m_1	m_1	m_3	m_2	e	r^2	r
m_2	m_2	m_1	m_3	r	e	r^2
m_3	m_3	m_2	m_1	r^2	r	e

Table 1: All possible element compositions.

Clearly we can see composition is closed, composition is associative, the identity transformation, and the inverse elements, which are the pairs of elements that become the identity element. So we say

$$D_3 = \{e, r, r^2, m_1, m_2, m_3\}$$

is a group under composition. We call this group the Dihedral group of order 3 (since triangles have 3 sides).

Notice that the table is not symmetrical. This is because, unlike our first example, the composition of transformations is not **commutative**. That is, the order in which we operate matters, for instance $rm_1 \neq m_1 r$. Groups can be commutative, like our first example, and are called **abelian** if so.

4.3 Permutation Groups

Why is knowing what a group is relevant to the Rubik's cube? Because, as we've extensively discussed already, the Rubik's Cube can be described using permutations. And it just so happens that permutations form groups too!

Let's look at an example of transformations of a triangle again. How does this have anything to do with permutations? Well, labelling each of the triangle's corners as before, we notice the transformations to the triangle are actually just permutations of the triangle's corners. Keeping track of this we can write the permutation associated with each transformation.

- Clearly the identity permutation will just be doing nothing to the triangle.
- Rotating the triangle simply cycles the corners about. These can be represented as (123) and (132).
- There are 3 mirror lines on the triangle. Each reflection just swaps two corners, while keeping the third in place. So the permutations describing reflections are (23), (12) and (13).

Hence the set of rotations of a triangle is the same as the set of these specific permutations of 3 points:

$$\{(), (123), (132), (23), (12), (13)\}$$

Since we know that rotations of a triangle form a group, these permutations must also form a group (with the same structure as the triangle rotations group). Check for yourself that this set with composition forms a group (you can assume permutation composition is associative).

This is actually the set of all possible permutations of 3 objects, often described as the symmetric group of order 3, or S_3. However, permutation groups do not need to contain every possible permutation to be a group, for instance the set of permutations of 4 objects $\{(), (12), (34), (12)(34)\}$ still forms a group under composition.

4.4 Group Properties

Knowing what rules a group must follow to be a group, we can prove some simple facts for any general group. These facts I won't go into proving, as then this video will turn into a course on abstract algebra, but I recommend looking at them if you're interested to see why these are true.

1. The identity of a group is unique.
2. The inverses of elements are unique.
3. Every element in a finite group has finite order. The order of an element, g is the smallest integer n such that $g^n = e$.
4. The order of an element will divide the size of the group.
5. A subgroup H of a group G is some set of elements of G that still forms a group under the same binary operation of G. If H is a subgroup of G, then the size of H divides the size of G.
6. We have cancellation laws, which are useful for simplifying equations. Say we have a group containing elements a and b with the relation $ba^2 = b$. Since b has an inverse, b^{-1}, we can multiply on the left by b^{-1}. Knowing $bb^{-1} = e$ and anything multiplied by the identity is itself, we're left with $a^2 = e$

To give you a taste, we'll show why the identity is unique.

Group also often have generators. The generators of a group $\mathcal{G}$ are sets of elements of $\mathcal{G}$ such that every element of $\mathcal{G}$ can be represented as a combination of these elements under the group's operation. For instance, the group of triangle transformations D_3 can be generated by just r and m_1. We represent this by writing $D_3 = \langle r, m_1 \rangle$.

5 The Rubik's Cube

5.1 The Rubik's Cube Group

Now we know that permutations can and do form groups, we can describe the Rubik's Cube Group!

Before we get ahead of ourselves though, let's actually check that the Rubik's Cube does actually form a group. Each element of our group will be the sequence of moves required to get to each state of the puzzle. Since we can describe each move as a permutation of the stickers of the puzzle, each element is just a really, really long permutation of the stickers. Then our binary operation will just be composition of these. We'll call this group $\mathcal{R}$.

Checking that $\mathcal{R}$ is indeed a group:

- Composition of moves are always elements of $\mathcal{R}$ by how we defined it. So $\mathcal{R}$ is closed.
- As we stated before, composition of permutations is associative. We can see this on the cube as well, since R (U F) is the same as (R U) F.
- The identity is just the element that applies no moves to the puzzle. (You can think of this as the solved state of the puzzle, but remember the elements of our group are actually the move sequences required to get to each state of the puzzle).
- Every move of the puzzle has an inverse, like R and R'. To inverse a sequence of moves, we follow the sequence of moves backwards, performing the inverse of each element, which can be clearly see on a normal Rubik's Cube. These inverse movements are obviously elements of $\mathcal{R}$, so $\mathcal{R}$ contains all necessary inverses.

The Rubik's Cube is a group!

It's important to remember that our group elements are not states the puzzle can be in, but the sequences of moves required to get to each state.

Another way we can think about $\mathcal{R}$ is what generates it. Since $\mathcal{R}$ is just all of the states the Rubik's Cube can be in, we can just say $\mathcal{R}$ is generated by all of the face turns of the cube. We represented this by saying

$$\mathcal{R} = \langle U, F, R, L, D, B \rangle$$

And remember, each turn of the puzzle is simply just a permutation of the pieces of the cube (or of the 48 stickers of the cube).

Using general group properties, we can see that every sequence of moves on the cube has an order. This means no matter what sequence of moves to make on a puzzle, if you repeat them over and over again, eventually you'll return to your starting position.

5.2 Size of the Rubik's Cube Group

Finding the size of the Rubik's Cube group will tell us how many possible combinations the Rubik's Cube has, as each element of $\mathcal{R}$ describes a state that the cube can be in.

5.2.1 The Screwdriver Group

Our first thought to calculate its size is to calculate the number of permutations the edge pieces and corner pieces have. We can do this using basic ideas of calculating the number of permutations of some number of objects. Let's imagine we completely disassemble the puzzle and are putting it back together piece by piece.

There are 12 edges on the puzzle, and each edge can equally occupy any edge position on the puzzle, hence there are 12! ways the edges pieces can be arranged on the cube. Each edge has 2 orientations it can be in, so we multiply by 2^{12} to account for the orientations.

There are 8 corner pieces on the puzzle, so we have 8! ways of arranging the corner pieces on the cube. Each corner piece has 3 different orientations, and so we multiply by 3^8 to account for the orientations.

All together we have $12! \times 2^{12} \times 8! \times 3^8 = 519,024,039,293,878,272,000$ combinations.

However, would all of these puzzle states be a part of $\mathcal{R}$? If a cube state is not an element of $\mathcal{R}$, we'll call the state 'unsolvable', since if it was in $\mathcal{R}$, then it would have an inverse that would 'solve' that state. So are all of the states we calculated 'solvable'?

5.2.2 Parity of the Rubik's Cube

Think back to even and odd permutations:

> A permutation is even/odd if it can be written as an odd/even number of 2-cycles.

If we look at the cycle notation of the U face, this is written as

$$(UF, UL, UB, UR)(UFR, UFL, UBL, UBR)$$

Which is just 2 disjoint 4-cycles. We know that 4-cycles have even parity, and even+even=even, hence every turn of the puzzle describes an even permutation. Thus, states of the puzzle that requires an odd permutation to get to must not be in the Rubik's Cube group $\mathcal{R}$ and hence not a solvable state.

This means a single swap of just two pieces is not a solvable state, nor is any odd number of piece swaps. This means, if you were to disassemble and reassemble the puzzle, only half of the time the puzzle would be solvable, neglecting the orientation of pieces. It's important to remember that, while the whole cube's parity is always even, it's possible for both the edge and corner parities separately to be odd (i.e. edges are an odd cycle and corners are an odd cycle), since odd + odd = even.

We haven't considered the orientation of the pieces. We'll look at edges and corners separately:

- We will show that the edges also share an orientation parity condition. But first we must define the 'orientation' of an edge when it is not in it's solved position. Define a pair of axes, starting at the centre of the puzzle, pointing out of each centre piece, and let each axis have a positive and negative direction (just like 3D coordinates). Now let the very edge of each edge piece have an arrow that points in the positive direction of the axis it points parallel to. Initially in the solved state, all edges points in the positive direction, and we define them all as being oriented. If we turn one face, say the F face, then the arrow configuration changes. Two of the edges remain pointing positively, while the other two point negatively, and hence become misoriented. Clearly only two edges become misoriented at a time, and no turning of other faces will re-oriented them (apart from the B face directly opposite the F face). Hence only an even number of edges can become misoriented. Therefore a single edge being misoriented is impossible. So if we were to randomly assemble the puzzle, half of the time it will be solvable.

- The corners have a different orientation condition than the edges. Take two opposite colours on the cube to be 'fancy' colours, for instance white and yellow, and hold them such that they face the top and bottom of the puzzle. The stickers on corner pieces with a 'fancy' colour will be marked as 0, the sticker immediately clockwise of them with 1, and the last face as 2 (the face immediately counter-clockwise from the 'fancy' colour). We'll call a corner oriented if a 'fancy' colour faces a 'fancy' side. In the solved state, if we count all of the numbers on the 'fancy' sides, we get 0. If we turn a 'fancy' side, no numbers change, hence the orientation of corners do not change. If we turn any other side, our numbers increase by 3 on both fancy sides, as one corner twists clockwise and another twists counter-clockwise. Therefore the change in the total number will always be a multiple of 3. This means that for any corner that is twisted clockwise, another corner must be twisted counter-clockwise. Since there is a 1/3 chance that our number is a multiple of 3, there's a 1/3 chance that when we assemble the cube randomly that it'll be solvable.

Clearly not all of these states are solvable, how many are? Rather than now trying to calculate the number of combinations again from scratch using these new conditions, it's much easier to think about the probability that the assembled cube is in a solvable state.

The probability that the puzzle is solvable is just $1/2 \times 1/2 \times 1/3 = 1/12$ of the time. So if we divide our initially calculated answer by 12, we get that the number of solvable/valid states of the Rubik's Cube is

$$\frac{12! \times 2^{12} \times 8! \times 3^8}{12} = 43,252,003,274,489,856,000$$

This is the size of the Rubik's Cube group!

This is so large, that if you were to draw each combination on a single sheet of paper, and stacked these papers, the stack of papers would reach Pluto 728 times.

5.3 Largest Order Element

We know that the elements of $\mathcal{R}$ are just the set of sequences of moves for every valid state the Rubik's Cube can be in. And we know as a general fact of groups that every element of $\mathcal{R}$ must have a finite order, i.e. if repeated a certain number of times, it will return to back to the original state. What is the maximum order an element from $\mathcal{R}$?

We know that the order of a permutation is just the lowest common multiple of the size of the cycles it's composed from, so we want to try and find the highest lcm cycles on the cube. We can achieve this by finding cycle lengths that don't share common factors, the best way we can guarantee this is by having the cycles have prime (or pure prime power) lengths.

This can be done through trial and error, but we find in the best case the edges can have a permutation with cycles length 4 and 7, and corners of length 3 and 5. This gives us permutations with orders 28 and 15 respectively. However, this isn't possible to do on the cube, since we know turns on the puzzle have even parity, and a permutation composed of cycles of lengths 3, 4, 5, 7 will have odd parity. This is mainly because all of our cycles have even parity except our 4 cycle which has odd parity. The smallest change we can make is splitting the 4-cycle into a pair of 2-cycles, which reduces the total order of the permutation by half, by now the permutation is back to being even![3] The order of our edges permutation is now 14, corners is still 15.

We haven't factored orientation of pieces into our order either! To increase it the most possible, we can say our edge permutation will flip an even number of edges after 14 repeats, doubling the order of the edge permutation to 28, and our corner permutation twists a corner clockwise and another counter-clockwise after 15 repeats, increasing the corner permutation 3-fold to 45.

So now the edge permutation has order 28, corner permutation has order 45. Hence

$$lcm(28, 45) = 28 \times 45 = 1260$$

To find an algorithm that has this order, we need to find a sequence of moves that: swaps two pairs of edges and cycles another 7 edges which leaves at least 1 edge flipped after 14 repeats; and has a 3 corner cycle and 5 corner cycle which leaves the sum of corner twists not equal to 0.

The algorithm R U2 D' B D' achieves this!

5.4 Subgroups (UNFINISHED)

We call a set A a subset of a set B, denoted $A \subset B$, if all of the element of A are also in B. So it's almost like A is contained in B. For instance, the set $\{2, 3, 5\}$ is a subset of $\{1, 2, 3, 4, 5, 6\}$.

We can extend this idea to groups: a subgroup $\mathcal{H}$ of a group $\mathcal{G}$ is a subset of $\mathcal{G}$ which is also a group, under the same binary operation as $\mathcal{G}$.

For example, the set of only rotations of a triangle (including the identity) is a subgroup of D_3 from earlier. It can be checked that this is indeed a group.

Subgroups are useful to look at, especially for the Rubik's Cube, as we can still learn about the puzzle with less complication.

5.5 Commutation

We've spent ages learning about how the Rubik's Cube is a group, but how can any of this help with finding a general solution to the puzzle? The first answer to this is called **commutation**.

We mentioned groups being *commutative* before; that's when the order in which we operate on elements in a group doesn't matter. Like in our example with the group of integers with addition, for any two integers a and b, clearly $a + b = b + a$, or $ab = ba$ in multiplication notation. This isn't true in general for groups, as we saw for the dihedral group D_3. We can rewrite this equation as $aba^{-1}b^{-1} = e$. We call the LHS the commutation of elements a and b, and sometimes write $[a, b] = aba^{-1}b^{-1}$.

In non-abelian groups, commutation of elements don't in general give us the identity. However, commutation can tell us 'how' commutative elements are to each other.

For example, let P=(1 2 3 4)(5 6 7) and Q=(5 7)(8 9 10 11 12) be permutations of 12 objects. The commutation $PQP^{-1}Q^{-1}$ isn't immediately clear, but we realise that since (1 2 3 4) is disjoint to all other cycles except its inverse in P^{-1}, which is just the inverse of the cycle, (1 2 3 4) just turns into the identity. Same goes for the (8 9 10 11 12) cycle. Hence $[P, Q] = PQP^{-1}Q^{-1} =$ (567)(57)(576)(75) = (756). So only the cycles that 'overlap' each other will be a part of the commutation. This is empirical and is not in general true.

[3] Since composing two odd parity cycles makes the overall parity even, odd+odd=even.

This would be very useful on the Rubik's Cube, as each turn is a long permutation, and it would help to be able to move a small number of pieces at a time, rather than 8 all at once. Let's try and make a commutation to cycle 3 corners (as a 2 corner cycle is impossible due to parity).

We'll let one of our commutator moves be a single face turn, say D. (we pick D as this makes commutators faster to execute). We want to only very slightly affect this turn, which we can think as 'overlapping' the permutations of the movements. The best way is to change one corner of the D face, for instance doing the moves R U' R'. We've replaced only one corner of the D face and nothing else; we've affected a lot of other pieces on the top layer, but these will resolve after the commutation since our D move has no effect on any of these other pieces. So we have
[R U' R', D] = (R U' R') D (R U R') D' is our commutator.

Performing this on the cube, as expected only a small number of pieces moved, 3 corner pieces precisely. The 3 corners that cycled are those that 'overlapped' during the commutation. This now gives us a method to cycle 3 corners where one is on the top face and two on the bottom face.

The exact same logic can be applied to the edges, we can use a middle layer turn such as M', as the equivalent to the D move for corners, and another set of moves that replaces one edge in the middle layer, like (U R U'). Then [U R U', M'] only cycles 3 edges, where two of them are in the middle layer.

We can also use this approach to flip/twist pairs of pieces (remember we can't flip singular pieces due to the cube's parity). This example we'll flip 2 edges in place, but the same principle can be applied to the corners as well.

Instead of replacing the middle layer with another random edge, let's take out a middle layer edge, then put it back but flipped. We can take out the DF edge from the D layer using M' U M (remembering to try and keep the D layer intact apart from the DF edge). To insert the edge back into the DF position flipped, we can do this by doing U' M' U2 M. We've now flipped the DF edge! We'll make our overlapping move D (or actually D' or D2 it doesn't actually matter as long as it's a different D layer edge). Now reversing our edge flipping moves will restore the messed up top layer and flip the new edge in the DF position. Reversing the last D move then re-aligns the bottom layer, leaving us with two flipped edges!

Using commutation, we're able to select any three pieces (edges or corners) on the puzzle, and cycle them: just pick three pieces such that a 3-cycle will solve at least one of them, and repeat until all pieces are in the correct places, but not necessarily oriented. We can then use commutators to orient two pieces at a time to solve the puzzle. Neat!

5.6 Conjugation

Commutation is great, but it has a fatal flaw. Given two sequences of moves X and Y, the commutation $[X, Y]$ always contains an even number of moves. This is because however many moves X and Y has, X^{-1} and Y^{-1} will have the same number of moves. This is problematic as it's possible for the edges and corners on the cube to separately have odd parity. Hence commutation would not be able to solve the cube state, and we'd be left with a single edge swap and corner swap.

How can we get around this? **Conjugation** is a process very similar to commutation, where we have two sequences of moves X and Y, and their conjugate is XYX^{-1}. There's a simple way to represent conjugation in terms of cycle notation.

Consider two permutations σ and τ, but let's think of them as functions. For instance if $\sigma = (1\,3\,4\,2)$, we can think of it a function that maps the set $\{1, 2, 3, 4\}$ to itself with the following properties:

$$\sigma(1) = 3, \sigma(2) = 1, \sigma(3) = 4, \sigma(4) = 2$$

This is actually where the original definition of a permutation comes from. The function σ is called a bijection, as it maps every element from $\{1, 2, 3, 4\}$ to itself, with a 1-1 correspondence.

Anyway, we can really nicely represent conjugation using this function notion. If σ and τ are some arbitrary permutations with $\sigma = (a\,b\,c)$, then the conjugation $\tau\sigma\tau^{-1} = (\tau(a)\,\tau(b)\,\tau(c))$. This is true because if we evaluate $\tau\sigma\tau^{-1}(\tau(a))$, the τ cancels, leaving us with $\tau\sigma(a)$, and we know $\sigma(a) = b$ due to the cycle we're given for σ. Hence the conjugation $\tau\sigma\tau^{-1}(\tau(a)) = \tau(b)$.

Why is any of this useful or relevant? Well on the cube, if we perform a quarter-turn on any side, we change the the corner/edge parity from even to odd or vice versa. This means if we have an odd edge or corner parity - which we can't solve with just commutators - we can perform a quarter turn to change it. How do we preserve the rest of the puzzle?

We can use conjugation to 'replace' a piece that would be cycled under a commutator. If we let X be some set-up and Y a commutator, then only the pieces which now take the positions of

the commutator's pieces will cycle. Hence we can resolve our corner-edge swapping situation and finally fully solve the puzzle.

If we have this J-perm to finish the cube off, which is a single corner swap and edge swap, we cannot solve using purely commutators. However, we can apply a U' to the puzzle and now we have a 3 corner and 3 edge cycle, both of which are solvable with a commutator (along with twisting/flipping them).

We have found our general solution!